Quantifying Brand Perception from Customer Reviews using Generative AI

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The Limits of Traditional Brand Perception Measurement

- Brand perception is **subjective and dynamic** – shaped by consumer expectations, emotions, and experiences. As such, there's **no single formula** to quantify it. Instead, organizations use a combination of: Structured surveys and sentiment scores, indirect proxies (like NPS).
- **Traditional surveys methods suffer low response rates**
  - According to this article by xola, average survey response rate is only **33%**.
  - According to delighted.com Net Promoter Score surveys typically see **<20%** response rate
- **Biases in traditional methods**
  - According to the AAPOR Task Force (2022), **social desirability bias** leads to **overreporting of good behavior.**
  - Combined with low participation (often **<15%**), this reduces **accuracy and representativeness.**
- **Surveys are slow and expensive**
  - **Conducting large-scale surveys is both costly and time-consuming**, making them less feasible for continuous tracking – especially at the scale of digital platforms.

Rethinking Brand Perception: The Approach

- **Consumers talk about brands all the time** – just not in surveys. Every product review, complaint, or compliment is a window into how people actually perceive a brand.
- These unstructured, spontaneous reviews are **rich with insight – but hard to analyze** at scale using traditional tools.
- This is where **Generative AI can help**. Since it can understand the **nuances of language**, it can extract **aspect-level sentiment** (e.g., battery, design, service), which we can then map to Philip Kotler's Three Levels of Product: **Core**, **Actual**, and **Augmented**.
- These categorized sentiments can then be **quantified into perception scores** (details in later slides), enabling **brand analysis** in a structured, scalable manner.

Rethinking Brand Perception: The Flow

Example Review
"The design feels cheap."

↓

LLM Model
+
Prompt

↓

Aspect : design,
Sentiment : negative
Type : actual

- Once the LLM processes a review, it extracts **aspects** (e.g., battery, design, support) and assigns:
- a **sentiment label** (positive, neutral, negative), and
- a **product level type** based on **Philip Kotler's Three Levels of Product**:
  - **Core** – fundamental benefit (e.g., performance, reliability)
  - **Actual** – tangible features (e.g., design, hardware quality)
  - **Augmented** – added value (e.g., customer service, accessories)
- Each aspect's sentiment is converted to a numeric value:
  - Positive → **+1**
  - Neutral → **0**
  - Negative → **–1**
- These values are averaged by product layer (**core, actual, augmented**) for each brand, resulting in **layer-wise perception scores**. The final **overall brand perception** is computed by averaging across all three layers.

Structure of the Presentation

- **Dataset**
- **Preprocessing**
- **Model Selection**
- **Prompt Engineering**
- **Perception Score Calculation**
- **Results**
- **Hypothesis Testing**
- **Conclusion & Future Scope**

Detailed Methodology: Step 1 – Dataset

- **Original Source**: https://amazon-reviews-2023.github.io/ (1996 -2023)
- **Data Format**: JSON structured review and metadata files.
- **Category**: Mobile Phones and Accessories
- **Scope**: Mobile phone reviews only
- **Volume**: ~9.34 GB of review data and ~4.02 GB of metadata.
- **Review Count**: 11.6M total reviews → filtered to 187,261 mobile phone reviews (2018–2023) after removing accessories and irrelevant items.
- **Geographic Scope**: United States only.
- **Analysis Timeframe**: 2018 to 2023.

Detailed Methodology: Step 1 – Dataset (contd.)

Field	Type	Explanation
rating	float	Rating of the product (from 1.0 to 5.0).
title	str	Title of the user review.
text	str	Text body of the user review.
images	list	Images that users post after they have received the product. Each image has different sizes (small, medium, large), represented by the small_image_url, medium_image_url, and large_image_url respectively.
asin	str	ID of the product.
parent_asin	str	Parent ID of the product. Note: Products with different colors, styles, sizes usually belong to the same parent ID. The "asin" in previous Amazon datasets is actually parent ID. **Please use parent ID to find product meta.**
user_id	str	ID of the reviewer
timestamp	int	Time of the review (unix time)
verified_purchase	bool	User purchase verification
helpful_vote	int	Helpful votes of the review

Data schema for reviews

Detailed Methodology: Step 1 – Dataset (contd.)

Field	Type	Explanation
main_category	str	Main category (i.e., domain) of the product.
title	str	Name of the product.
average_rating	float	Rating of the product shown on the product page.
rating_number	int	Number of ratings in the product.
features	list	Bullet-point format features of the product.
description	list	Description of the product.
price	float	Price in US dollars (at time of crawling).
images	list	Images of the product. Each image has different sizes (thumb, large, hi_res). The "variant" field shows the position of image.
videos	list	Videos of the product including title and url.
store	str	Store name of the product.
categories	list	Hierarchical categories of the product.
details	dict	Product details, including materials, brand, sizes, etc.
parent_asin	str	Parent ID of the product.
bought_together	list	Recommended bundles from the websites.

Data schema for metadata

Detailed Methodology: Step 1 – Dataset (contd.)

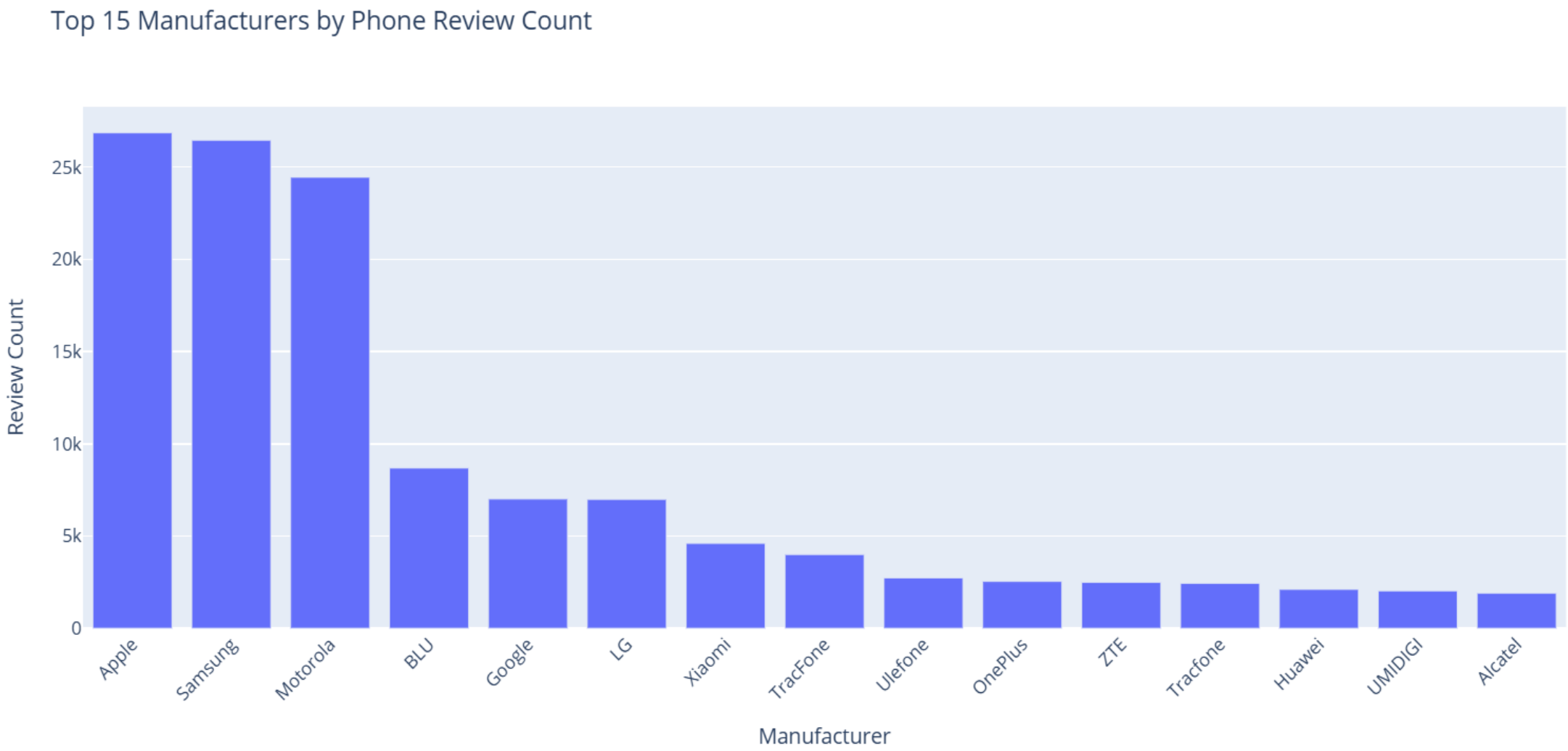

```
......
{
  "sort_timestamp": 1634275259292,
  "rating": 3.0,
  "helpful_votes": 0,
  "title": "Meh",
  "text": "These were lightweight and soft but much too small for my liking. I would have preferred
two of these together to make one loc. For that reason I will not be repurchasing.",
  "images": [
    {
      "small_image_url": "https://m.media-amazon.com/images/I/81FN4c0VHzL._SL256_.jpg",
      "medium_image_url": "https://m.media-amazon.com/images/I/81FN4c0VHzL._SL800_.jpg",
      "large_image_url": "https://m.media-amazon.com/images/I/81FN4c0VHzL._SL1600_.jpg",
      "attachment_type": "IMAGE"
    }
  ],
  "asin": "B088SZDGXG",
  "verified_purchase": true,
  "parent_asin": "B08BBQ29N5",
  "user_id": "AEYORY2AVPMCPDV57CE337YU5LXA"
}
......
```


Sample review

Detailed Methodology: Step 2 – Preprocessing

- Initial dataset: 11.6M reviews from Cell Phones & Accessories (1996–2023)→ Filtered to 187,261 mobile phone reviews (2018–2023)
- **Preprocessing tools:**
  - Downloaded via Hugging Face Datasets library
  - Apache Spark used for scalable joins and filtering, meta_data and reviews joined using shared key: parent_asin
- **Filtering**
  - Used category tree (last_category, second_last_category) to remove accessories, mounts, etc. categories=['Cell Phones & Accessories', 'Cases, Holsters & Sleeves', 'Basic Cases']
  - Used zero-shot classifier (BART) to identify product type. Only retained reviews confidently labeled as Mobile Phones
- **Metadata Enrichment**
  - Cleaned and standardized brand names
  - Focused on top 15 brands by count
- **Text Cleaning**
  - Removed HTML tags, emojis, redundant punctuation. Filtered out reviews with <5 words.

Detailed Methodology: Step 2 – Preprocessing



Detailed Methodology: Step 3 – Model Selection

Model	Type	Cost	Accuracy (Sentiment)	Notes
GPT-4	Paid	~$30 per 1M tokens	Very high (few-shot needed)	Best quality; slowest and most expensive
GPT-3.5-turbo	Paid	~$1.5 per 1M tokens	Good	Cost-effective for short reviews; ~20× cheaper than GPT-4
Claude 3 Haiku	Paid	~$0.50 per 1M tokens	Good to very good	Fast and slightly cheaper than OpenAI
Cohere Command R+	Free-tier & Paid	Free up to a limit	Good	Accurate with strong instruction following
Mistral-7B Instruct	Open-source (local)	Free (infra cost only)	Moderate–High	Needs ≥16GB GPU; good accuracy
Mixtral-8x7B	Open-source (local)	Free (infra cost only)	Very high (best open model)	Needs 32–48GB GPU; state-of-the-art for local setup
LLaMA 2–7B Chat	Open-source (local)	Free (infra cost only)	Good (general)	Lightweight; works for basic sentiment
LLaMA 3.3–70B (Used)	Open-source + Groq	Free (Groq API credits)	Very high + scalable	Chosen for this project: balanced cost, accuracy, and high throughput

Model selection analysis

Detailed Methodology: Step 4 – Prompt Engineering

- A custom **prompt was crafted** to extract **product aspects**, determine **sentiment polarity**, and classify them into **perception layers**:
  - **Core, Actual, Augmented**
  - The LLM also inferred whether the **core promise** of the product was:
  - **Fulfilled**, **Unfulfilled**, or **Unclear**
- Output returned as structured **JSON format**, enabling:
  - easy downstream parsing
  - Quantitative scoring and benchmarking across brands

Detailed Methodology: Step 4 – Prompt Engineering (contd.)

```
def build_prompt(text):
    return f"""
You are given a customer product review.

Your tasks:
1. Extract all product aspects mentioned in the review.
2. For each aspect, determine:
   a. Sentiment: "positive", "negative", or "neutral"
   b. Type: whether the aspect relates to the "core" product or the "actual" product or the augmented product

Definitions:
The concept of a core product originates from Philip Kotler, in his 1967 book – Marketing Management: Analysis, Planning and Control.
It forms the first level of the concept of Three Levels of a Product.
Kotler suggested that products can be divided into three levels: core product, actual product and augmented product.
The core product is defined as the benefit that the product brings to the customer. The actual product refers to the tangible object and relates to the physical quality and the design.
The augmented product consists of the measures taken to help the consumer put the actual product to use. By using a mixture of the three levels
of product in research and development, business firms can better understand their customers, better position themselves in the market and create a more successful product.

3. Based on the aspects extracted, determine if the review reflects that the product fulfilled its core promise.
   Return one of the following:
   - "fulfilled": if the core aspects are mostly positive or clearly aligned with the brand's promise (for brand promise, use your own knowledge)
   - "unfulfilled": if core aspects are mostly negative or show a disconnect
   - "unclear": if core aspects are not mentioned or are ambiguous

Return your answer in this exact JSON format (ensure that json is valid even incase of reviews where extraction is unclear):

{{
  "aspects": [
    {{"aspect": "...", "sentiment": "...", "type": "..."}},
    ...
  ],
  "core_alignment": "fulfilled"  // or "unfulfilled", or "unclear"
}}

Return only the valid JSON and nothing else.

Review:
{text}
""".strip()
```

Prompt template for aspect extraction

Detailed Methodology: Step 4 – Prompt Engineering (contd.)

I returned this and bought the US model from AT&T. I was happy with the phone until it fell and the screen cracked. Samsung claims to have Gorilla Glass 6 which is supposed to withstand a drop of 1.5 meters. It slipped out of my pocket from a seated position at less than one-third that distance. It also had a Samsung case & screen protector, and fell onto leaves before hitting the ground. Samsung Support does not stand behind their products either and described the glass as "brittle." Their words, not mine.

In general, I don't like the curved edges. The curve makes for more surface area susceptible to impact which many cases don't adequately protect. It's also impossible to find a good tempered glass screen protector—either it has a circle for the front fingerprint sensor or the sensor doesn't work at all. This is not the case for flat phones with front fingerprint sensors. The curve makes it hard to draw on or see the edges when framing a photograph. Sadly, Samsung is the only phone with an EMR stylus.

Original Review:
This international model with dual SIM was supposed to work with American SIMs but customer service at AT&T said they couldn't activate this phone because they're phasing out 3G. I ended up getting the US model through AT&T instead. Sucks that the US model only has one SIM even though it has space for a second.

If this works with your service provider, then the Note 10 is a great phone. Many see this as a downgrade from Note 9 because of the lower battery capacity of 3500mAh, but it still gets me through most of the day with some to spare. It's disappointing that the SD card and headphone jack were removed, but I wasn't using either on my Note 9 since there's plenty of space and I mostly use BT headphones. The phone also comes with USB-C headphones. The camera array includes a third wide-angle lens which is a significant upgrade. Although it's only slightly smaller than the 9, it is more pocketable. The smallest of the modern Note series—and flagship phones in general—are getting too big nowadays.


```
{
  "aspects": [
    { "aspect": "Gorilla Glass 6", "sentiment": "negative", "type": "actual" },
    { "aspect": "screen durability", "sentiment": "negative", "type": "actual" },
    { "aspect": "curve edges", "sentiment": "negative", "type": "actual" },
    { "aspect": "tempered glass screen protector compatibility", "sentiment": "negative", "type":
    { "aspect": "fingerprint sensor", "sentiment": "neutral", "type": "actual" },
    { "aspect": "EMR stylus", "sentiment": "positive", "type": "actual" },
    { "aspect": "dual sim functionality", "sentiment": "negative", "type": "actual" },
    { "aspect": "3G phase-out compatibility", "sentiment": "negative", "type": "augmented" },
    { "aspect": "battery capacity", "sentiment": "neutral", "type": "actual" },
    { "aspect": "SD card removal", "sentiment": "negative", "type": "actual" },
    { "aspect": "headphone jack removal", "sentiment": "negative", "type": "actual" },
    { "aspect": "USB-C headphones", "sentiment": "positive", "type": "augmented" },
    { "aspect": "camera array", "sentiment": "positive", "type": "actual" },
    { "aspect": "pocketability", "sentiment": "positive", "type": "actual" },
    { "aspect": "Samsung Support", "sentiment": "negative", "type": "augmented" }
  ],
  "core_alignment": "unfulfilled"
}
```


Sample review and extracted aspects

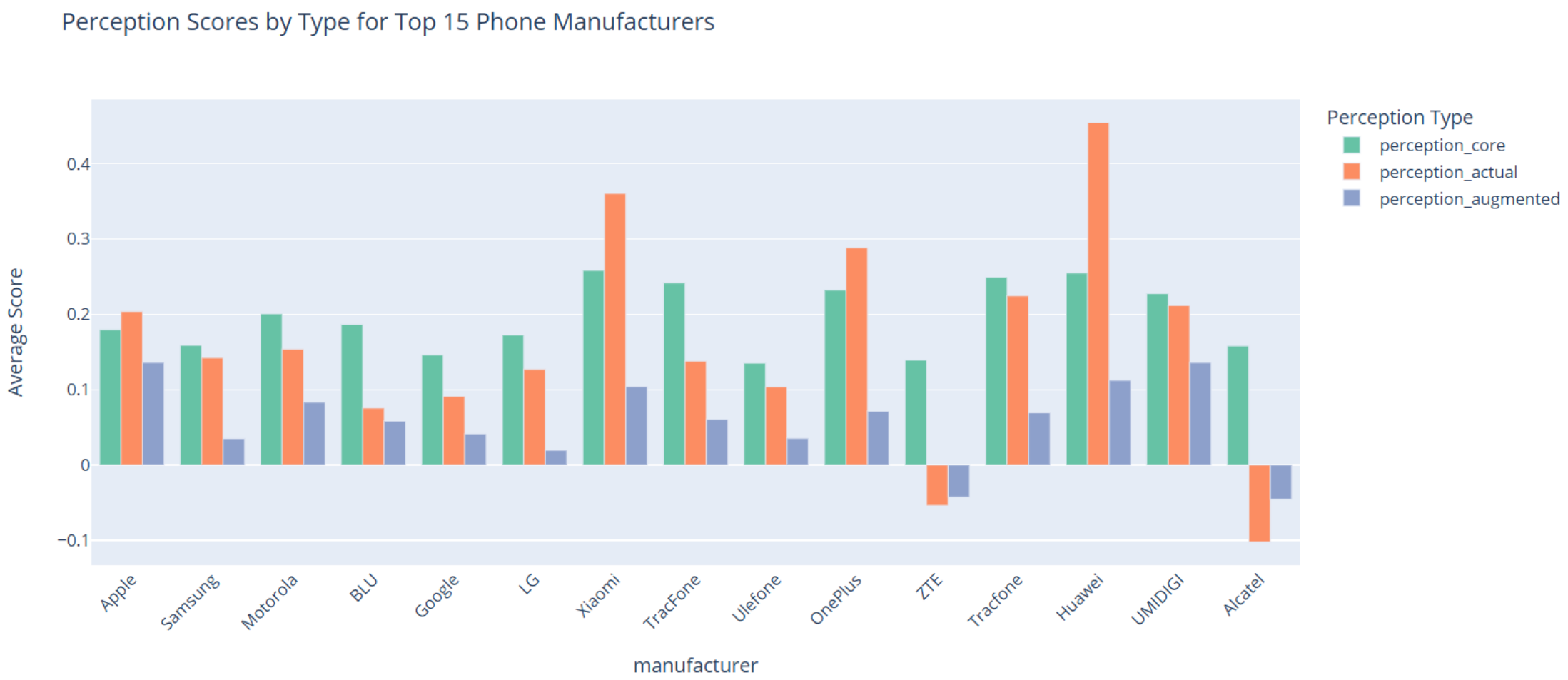
Detailed Methodology: Step 5 – Score Calculation

- We calculate perception scores separately for each product layer – **core**, **actual**, and **augmented** – by assigning numerical values to sentiment:
  - Positive → +1
  - Neutral → 0
  - Negative → –1
- **Actual Product Aspects**
  - Gorilla Glass 6 (–1)
  - Screen durability (–1)
  - Curve edges (–1)
  - Tempered glass screen protector compatibility (–1)
  - Fingerprint sensor (0)
  - EMR stylus (+1)
  - Dual SIM functionality (–1)
  - Battery capacity (0)
  - SD card removal (–1)
  - Headphone jack removal (–1)
  - Camera array (+1)
  - Pocketability (+1) →
- **Total Score: –4 → Count: 12**
- **Perception Due to Actual Product Aspects = –4 / 12 = –0.33**

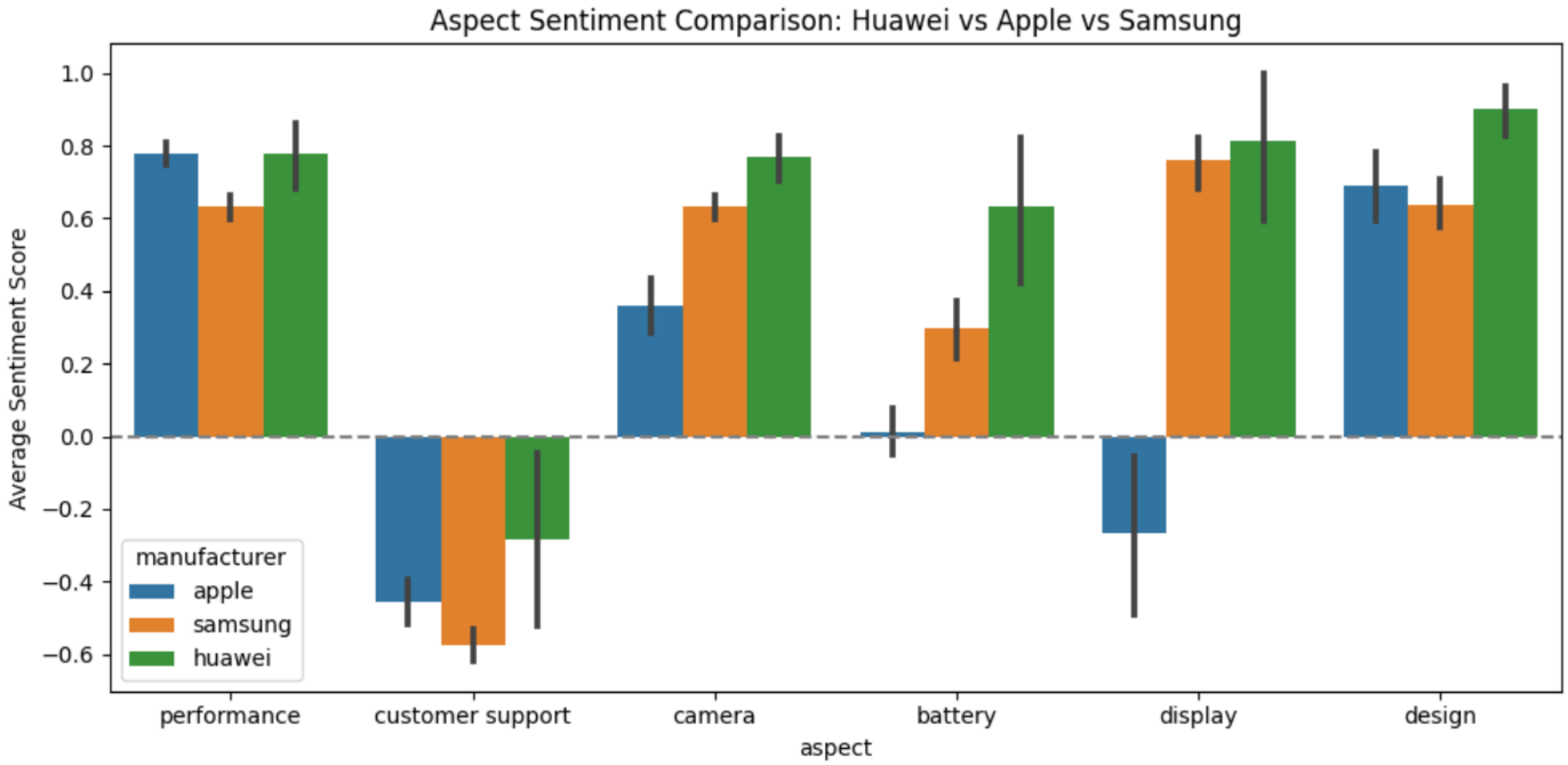

```
{
  "aspects": [
    { "aspect": "Gorilla Glass 6", "sentiment": "negative", "type": "actual" },
    { "aspect": "screen durability", "sentiment": "negative", "type": "actual" },
    { "aspect": "curve edges", "sentiment": "negative", "type": "actual" },
    { "aspect": "tempered glass screen protector compatibility", "sentiment": "negative", "type":
    { "aspect": "fingerprint sensor", "sentiment": "neutral", "type": "actual" },
    { "aspect": "EMR stylus", "sentiment": "positive", "type": "actual" },
    { "aspect": "dual sim functionality", "sentiment": "negative", "type": "actual" },
    { "aspect": "3G phase-out compatibility", "sentiment": "negative", "type": "augmented" },
    { "aspect": "battery capacity", "sentiment": "neutral", "type": "actual" },
    { "aspect": "SD card removal", "sentiment": "negative", "type": "actual" },
    { "aspect": "headphone jack removal", "sentiment": "negative", "type": "actual" },
    { "aspect": "USB-C headphones", "sentiment": "positive", "type": "augmented" },
    { "aspect": "camera array", "sentiment": "positive", "type": "actual" },
    { "aspect": "pocketability", "sentiment": "positive", "type": "actual" },
    { "aspect": "Samsung Support", "sentiment": "negative", "type": "augmented" }
  ],
  "core_alignment": "unfulfilled"
}
```


Extracted aspects and score calculation

Detailed Methodology: Step 6 – Results

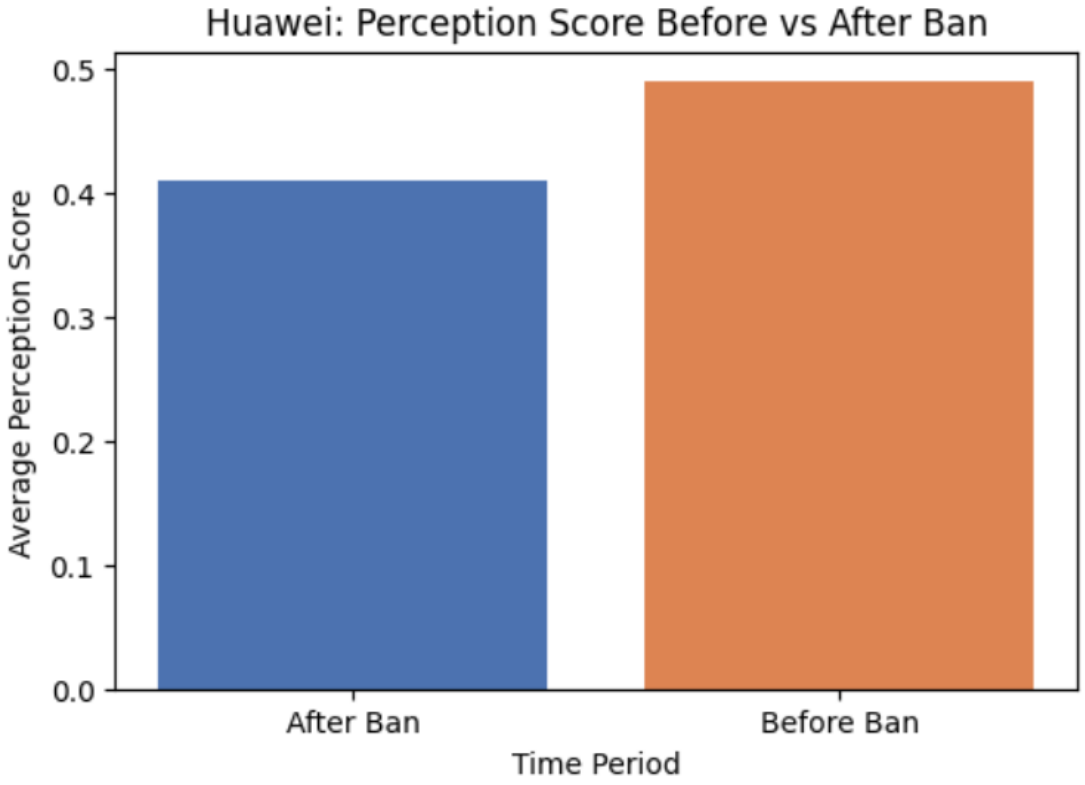


Detailed Methodology: Step 6 – Results (contd.)



Zooming In: Huawei and the U.S. Ban

- Huawei faced growing scrutiny from the U.S. and allies over suspected **backdoors** in telecom equipment, potentially enabling Chinese government surveillance.
- **Aug 2018: NDAA Bans Federal Use**
  The **National Defense Authorization Act (FY2019)** banned U.S. government agencies from using Huawei/ZTE hardware, citing cybersecurity concerns.
- **May 2019: Entity List & Executive Order**
  Huawei was added to the U.S. Department of Commerce Entity List, blocking it from doing business with American firms (e.g., Google, Intel, Qualcomm).
- **Underlying Allegations Behind the Ban**
  - Espionage & surveillance risks
  - Use of outdated, insecure firmware (e.g., VxWorks, telnet vulnerabilities)
  - Alleged ties to China's military & human rights abuses (e.g., Uyghur labor)
  - Violations of U.S. sanctions via tech exports to Iran



Hypothesis Testing – Understanding the Full Picture

- In earlier slides, we noted that **Huawei demonstrated comparatively elevated average perception ratings** and robust sentiment across various product dimensions.
- These findings necessitated a comprehensive statistical analysis to ascertain if Huawei's perceived dominance is consistent across perception dimensions and if its post-ban perception exhibits statistically significant differences.
- I aim to evaluate key differences in brand perception across the population using hypothesis testing – specifically:
  - **Before vs. After the Huawei U.S. Ban**
  - **Across Brands**, with a focus on **Huawei vs. Apple and Samsung**
- All tests assume a **significance level of $\alpha = 0.05$**, and results are interpreted under the assumption that the reviews analyzed are **representative of the U.S. population**.

Hypothesis Testing – Test 1 : Huawei Perception Before vs After the U.S. Ban

- **Hypotheses**:
  - H_0: There is no difference in Huawei's average perception score before and after the U.S. ban.
  - H_1: There is a statistically significant difference in Huawei's average perception score before and after the ban.
- **Test Used**: Independent two-sample t-test
- **Significance Level**: 0.05
- **Result**:
  - t-statistic = 2.5776
  - p-value = 0.0101
- **Interpretation**: We **reject the null hypothesis**. There is a **statistically significant difference** in Huawei's average perception score **before and after the U.S. ban**. This suggests that the ban **had a measurable impact** on how people perceive Huawei. (Since t statistic is positive, we can say that **mean perception before ban was greater than after ban**)

Hypothesis Testing – Test 2 : Overall Perception Differences Across Brands

- **Hypotheses**:
  - H_0: All brands have equal average perception scores.
  - H_1: At least one brand differs.
- **Test Used**: One-way ANOVA with Tukey's HSD
- **Significance Level**: 0.05
- **Result**:
  - F-statistic = 71.68
  - p-value = 1.98e-204
- **Interpretation**: We reject the null hypothesis. There are significant differences in Overall brand perception across manufacturers.

Hypothesis Testing – Test 2 : Overall Perception Differences Across Brands (contd.)

- **Post Hoc Insights**:
  - Apple scored significantly higher than Huawei
    (mean diff = 0.0976; $p < 0.01$)
  - Samsung scored significantly higher than Huawei
    (mean diff = 0.1617; $p < 0.01$)
  - Assuming that review data is representative of the U.S. population, the results from the Tukey HSD test imply: **Samsung > Apple > Huawei** in terms of mean overall brand perception

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1  group2 meandiff p-adj  lower   upper  reject
-----------------------------------------------------
 apple  huawei   0.0976   0.0  0.0734  0.1218   True
 apple samsung  -0.0641   0.0 -0.0743 -0.0539   True
huawei samsung  -0.1617   0.0 -0.1858 -0.1376   True
-----------------------------------------------------
```

Conclusion and Future Scope

- Developed a **scalable, interpretable framework** for quantifying brand perception using **unstructured customer reviews**.
- Processed **187,000+ U.S. mobile phone reviews**, combining **Generative AI**, **aspect-based sentiment extraction**, and **statistical testing**.
- Achieved three core objectives:
  - Extracted **brand-level perception** directly from raw reviews.
  - Differentiated between **core**, **actual**, and **augmented** perception layers.
  - **Compared brands** (e.g., Huawei vs. Apple) and **validated perception differences** statistically.
- **Limitations**
  - **Geographic Scope**: U.S.-only data; may not reflect global consumer behavior.
  - **Model Sensitivity**: Results depend on LLM choice and prompt quality (LLaMA 3.3–70B used here).

Conclusion and Future Scope (contd.)

- **Future Scope**
  - **Perception–Sales Correlation**:
    Investigate how perception scores correlate with actual sales metrics (e.g., quarterly shipments).
  - **Temporal Sentiment Tracking**:
    Track brand perception over time to assess impact of **product launches, controversies, or marketing campaigns**.

Thank You !